STACKS, DISCRIMINANTS: LEMMA-DUALIZING-COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 464-469 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 471-490, the complete author proof (18 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 36 of queue_discriminant.json (pb=634, stmt 464-469, proof 471-490).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Lemma 0.1. *Let $A \to B \to C$ be quasi-finite homomorphisms of Noetherian rings. There is a canonical map $\omega_{B/A} \otimes_B \omega_{C/B} \to \omega_{C/A}$.*

Proof. Choose $A \to B' \to B$ with $A \to B'$ finite such that $\mathrm{Spec}(B) \to \mathrm{Spec}(B')$ is an open immersion. Then $B' \to C$ is quasi-finite too. Choose $B' \to C' \to C$ with $B' \to C'$ finite and $\mathrm{Spec}(C) \to \mathrm{Spec}(C')$ an open immersion. Then the source of the arrow is

$$\mathrm{Hom}_A(B', A) \otimes_{B'} B \otimes_B \mathrm{Hom}_B(B \otimes_{B'} C', B) \otimes_{B \otimes_{B'} C'} C$$

which is equal to

$$\mathrm{Hom}_A(B', A) \otimes_{B'} \mathrm{Hom}_{B'}(C', B) \otimes_{C'} C$$

This indeed comes with a canonical map to $\mathrm{Hom}_A(C', A) \otimes_{C'} C = \omega_{C/A}$ coming from composition $\mathrm{Hom}_A(B', A) \times \mathrm{Hom}_{B'}(C', B) \to \mathrm{Hom}_A(C', A)$. □

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